

Student Guide Working with JupyterLite

Welcome! In this course you will write and run Python directly **in your web browser**, using a tool called **JupyterLite**. There is **nothing to install**, **no account to create**, and **nothing to download**: when you open the course link, all the notebooks and data files are already there. This short guide tells you everything you need. Read it once before the first lesson.

The one link you need

Course home: <https://andreatagliabue.github.io/python-unige/>

That single link opens **everything** all the lessons and the data are already there. When it loads, you will see a file browser on the left: just **double-click the lesson** you want (e.g. `01_introduction.ipynb`) and it opens in the center.

Tip: bookmark this link. Always open it in a **Private / Incognito** window (see Section 1 for why).

1. What is JupyterLite (and what you need)

- **JupyterLite** is a version of the Jupyter Notebook environment that runs entirely inside your browser. Python itself runs on your computer, *inside the browser tab* not on a remote server.
- **You do NOT need to install Python, Anaconda, or anything else.**
- **You do NOT need to log in or create an account.**
- **You do NOT need to upload anything:** the course materials are already loaded for you.

What you DO need:

- A laptop (Windows, macOS, or Linux it does not matter).
- A recent version of **Google Chrome, Microsoft Edge, or Firefox**. Avoid Internet Explorer and very old browsers.
- An **internet connection**, especially the first time you open it (the browser downloads Python and the scientific libraries the first time this can take 3060 seconds).
- Always open the course in a **Private / Incognito** window (Chrome: `Ctrl+Shift+N` / `Cmd+Shift+N`; Firefox: `Ctrl+Shift+P`). This ensures you always start from a clean, fresh environment no leftover state from a previous session. The tradeoff is that your

edits are not saved when you close the window; download your notebook before closing if you want to keep it (see Section 5).

2. Opening the course

1. Open your browser.
2. Go to the **course home**:

<https://andreatagliabue.github.io/python-unige/>

1. Wait a few seconds. You will see the **JupyterLab interface**:
2. on the **left**, a *file browser* showing the course notebooks and a **data** folder (these are already there for you);
3. in the **center**, the working area where notebooks open as tabs.

First time is slower. The very first time you run Python code, the browser downloads the Python engine and libraries. This happens only once per browser; afterwards it is fast.

3. Opening and running a notebook

1. In the **left file browser**, **double-click** the notebook for the lesson (a file ending in `.ipynb`). It opens as a tab in the center.
2. A notebook is a sequence of **cells**. There are two kinds:
3. **text cells** (explanations, in formatted text);
4. **code cells** (grey boxes containing Python).
5. To **run a cell**: click on it, then press **Shift + Enter**.
6. The code runs, the result appears just below the cell, and the selection moves to the next cell.
7. The small number in brackets to the left of a code cell, e.g. `[1]`, shows the order in which cells were run. While a cell is running you see `[*]`.

Run everything in order

The **order matters**: later cells often depend on earlier ones. The safest way to start a lesson is:

- Open the menu **Run** → **Run All Cells** to execute the whole notebook from top to bottom.

If something behaves strangely, reset and start clean:

- **Kernel** → **Restart Kernel and Run All Cells** this clears everything and runs the notebook again from the beginning.

Golden rule for beginners: if you get confused, do **Restart Kernel and Run All Cells**. Most "it doesn't work" problems disappear after this.

4. Editing and experimenting

- Click inside a code cell to edit the Python text, then press **Shift + Enter** to run it again.
 - Add a new cell with the **+** button in the toolbar.
 - It is completely normal to get **error messages** (red text). An error does not mean you are bad at this it just means Python found something it could not understand. Read the **last line** of the error: it usually tells you what went wrong.
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5. Saving your work (please read this is important!)

Because you are working in **Private / Incognito** mode, **your edits exist only for the duration of that window**. When you close the incognito window (or all incognito windows), the browser discards everything your changes are gone.

Therefore, if you want to keep your work, download it before closing:

1. Open the menu **File** → **Download** (or right-click the file in the left browser → **Download**).
2. Save the **.ipynb** file somewhere safe on your computer.

If you want to continue from a downloaded notebook in a later session, drag it into the file browser to upload it.

i This is intentional: starting each lesson in a fresh incognito window means you always get the original, unmodified course materials with no leftover state from previous sessions.

6. Troubleshooting

Problem	What to do
NameError: name '...' is not defined	You probably ran cells out of order, or skipped one. Do Kernel → Restart Kernel and Run All Cells .
First cell takes very long / seems stuck	The first run downloads Python and the libraries. Wait up to a minute and make sure you have internet.
A cell shows [*] and never finishes	

Problem	What to do
	It may be waiting or stuck. Use Kernel → Interrupt Kernel , then run again.
Everything looks broken (errors everywhere)	This is almost always the <i>running state</i> , not your file. Do Kernel → Restart Kernel and Run All Cells reloading the page also resets the running state. This fixes the large majority of issues.
I edited the notebook and want the ORIGINAL back	Close all incognito/private windows (not just the tab the whole incognito session), then open a new incognito window and reopen the course link. This resets the browser storage completely and loads the original files.
FileNotFoundError when reading a data file	Make sure you are running the notebook from the course link (where the <code>data</code> folder is provided). If it persists, reload the page and tell the instructor.

7. Quick reference

- **Run a cell:** Shift + Enter
- **Run the whole notebook:** Run → Run All Cells
- **Clean restart:** Kernel → Restart Kernel and Run All Cells
- **Save to your computer:** File → Download
- **Add a cell:** + button in the toolbar

You are ready. See you in the first lesson!